BAYTREE
ENGINEERING & CONSTRUCTION

Baytree Engineering & Construction Ltd.

Engineering Tomorrow's Infrastructure – Durable, Sustainable, Unmatched.

BAYTREE BUSINESS Profile

100%

Brochure **Product Catalogue**

Website
www.baytreeng.com

Executive Summary

Baytree Engineering & Construction Limited delivers specialised infrastructure across **marine and coastal works, public works and civil engineering, and integrated construction solutions.**
Formed in 2023 from the merger of Ako Adeniyi Ltd and allied entities, Baytree combines decades of execution experience with modern project management, advanced geosynthetics, and a client-focused delivery model.
Leadership you can trust. Baytree is led by **Bisola Smith**, an engineering executive with more than 28 years of experience across design, planning, erosion and flood control, and program delivery. He is an alumnus of **Lagos Business School, SMP64**, with technical certifications in erosion control, streambank protection design, operations management, and ISO 14001 and 9001 implementation. This background makes him a true hybrid leader, equally effective on-site and in the boardroom.

100%

What we deliver.

- **Marine and Coastal Infrastructure:** shoreline defence, revetments, seawalls, waterfronts, and erosion control.
- **Public Works and Civil Engineering:** roads, drainage, bridges, civic facilities, and river training.
- **Integrated Construction Solutions**: structural development and engineered building supply with installation and commissioning.

Proven results.

- **Bridge pier abutment project (Delta State Government/DESOPADEC) at Uzere, Isoko, Delta State,** using gabions, Terraforce ACBM (Terracrete) and geotextile to resolve/rectify a collapsed section of the bridge, as it was impassable.
- **Several soil stabilisations for NGIC pipelines across the country** – Agbede (1), Fugar (1), Uromi (2) in Edo State and Okpangede (2) and Power Station (1) in Kogi.
- **ACBM for HARTLAND LTD** on their NEWMAP Project in Benin
- **ACBM for EMMYBLESS NIG LTD** at Isele-Uku on their NEWMAP (Nigeria Erosion Watershed Management Program)
- **Coastal Protection & Beach Restoration:** our team delivered an end-to-end coastal protection solution—offshore breakwaters, geotextile tubes, and concrete block systems—drawing on hands-on experience in similar projects. We achieved rapid beach accretion while ensuring shoreline stability and preserving community access, from project acquisition through full execution.

Executive Summary

Proven results.

Baytree Engineering & Construction Ltd, together with our partners and OEM suppliers, brings a rich track record of successful projects across Africa and beyond. The following highlights represent our combined expertise and delivery capability.

- **Nelatur, Krishnapatnam, Andhra Pradesh:** ground improvement for a power plant ash pond bund raised by five meters on weak fly ash using geocells, biaxial geogrids, and nonwoven geotextiles. The solution increased bearing capacity, reduced settlement risk, and eliminated the need for deep foundations.
- **Orsang River, Katol, Gujarat:** reinforced soil wall with gabion fascia over a 184-meter stretch with bank heights up to 9.6 meters, including a launching apron to resist scour. The design stabilised the riverbank and enabled land reclamation with a flexible, permeable system that blends into the environment.

Technology advantage.

Baytree integrates world-class products through partnerships with **Techfab India**, **Inland Tarp and Liner**, and **Terraforce**. Solutions such as fabric-form concrete mattresses, geotextile tubes and bags, geocells, geogrids, and modular retaining systems deliver durability, speed, and sustainability with lower whole-life cost.

Quality, safety, and sustainability. We operate ISO-aligned processes, enforce zero-incident safety goals, and design for long life and low maintenance. Environmental stewardship is embedded through materials selection, drainage-enabled linings that relieve hydrostatic pressure, and designs that allow natural vegetation to reinforce stability.

Financial capacity and risk control. Baytree maintains adequate capital reserves, resilient supply chains, and OEM-backed logistics. We apply structured risk identification, quantitative analysis, and contingency planning to protect time, cost, and quality from day one.

Transparent value. Our pricing approach is tailored to the scope and site, fully itemised, and engineered for lifecycle value rather than the lowest first cost. Clients gain measurable performance, reduced maintenance, and assured delivery.

Recognition and industry engagement:

Leadership recognition includes the NICE Service and Contribution Award, as well as speaking roles at WAPI. Training at Flint USA on the use of coastal tubes and marine mattresses for coastal erosion and associated works, Geo Products USA for geocell manufacturing and application, and World of Concrete (WOC) USA for technology updates/transfer.

Mission Statement: Delivering specialised marine infrastructure, public works, and integrated construction solutions for government, municipal, and private sector clients.

Company Background and Credentials

Baytree Engineering & Construction Limited is a forward-looking infrastructure solutions provider, delivering specialised expertise across marine and coastal infrastructure, public works, and integrated construction solutions. Established in 2023 through the merger of Ako Adeniyi Ltd (founded in 2009) and other strategic entities, Baytree combines decades of hands-on engineering experience with modern project management and innovation-driven practices.

From conceptualisation to completion, we take a **client-focused, results-driven approach**—providing solutions that not only meet technical specifications but also stand the test of time. Our reputation is built on **quality, integrity, environmental stewardship, and collaborative delivery**.

Our Expertise

Marine & Coastal Infrastructure

Specialised shoreline engineering and defence solutions, including seawalls, erosion control, waterfront developments, and coastal protection systems.

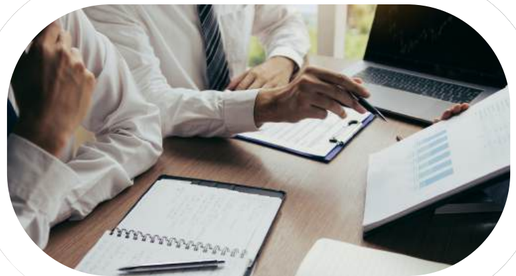
Public Works & Civil Engineering

Comprehensive municipal and civic infrastructure projects, including roads, drainage systems, bridges, and public facilities.

Integrated Construction Solutions

End-to-end structural development and building supply services, supporting major projects with construction, installation, and supply chain integration.

Fully Packaged Technical Proposals



Every engagement comes with comprehensive technical documentation, detailed drawings, and cost-benefit analyses, enabling clients to make informed decisions backed by data and precision engineering.

Company Background and Credentials

Our combination of technical depth, emergency readiness, and supply chain advantage ensures our clients enjoy infrastructure solutions that are faster, stronger, and more sustainable than industry norms.

Our Expertise

Rapid Emergency Response

In urgent situations such as flooding, shoreline erosion, or collapsed roads, BTE mobilises faster than any competitor, deploying ready-to-install, durable solutions that restore safety and functionality in record time.

OEM-Backed Speed Advantage

Our exclusive partnerships with global Original Equipment Manufacturers mean we can deliver results up to 60% faster than competitors, thanks to ready stock availability and streamlined supply chains.

Technical Mastery Across Sectors

From eroded shoreline restoration to industrial embankment stabilisation and advanced slope protection systems, we integrate proven engineering practices with cutting-edge technology for lasting results.

Mission Statement:

Delivering specialised marine infrastructure, public works, and integrated construction solutions for government, municipal, and private sector clients.

OEM-Sourced

And Certified Quality/Manufacturing

Through partnerships with **Techfab India**, **Inland Tarp & Liner**, and **Terraforce**, we provide products that meet international performance standards and are recognised in technical specifications globally. This ensures **consistent quality**, **fast availability**, and **full technical support** from design to installation.

- TECHFAB India
- Terraforce
- Inland Tarp and Liner

Leadership Team:

CEO/MD – Bisola SMITH.

At the helm of Baytree is **Bisola SMITH**, a highly respected industry leader with over **28 years of professional experience** in engineering, construction, civil works, flood and erosion control, and large-scale project management.

From the MD/CEO's Desk

At **BAYTREE ENGINEERING AND CONSTRUCTION LTD**, we deliver innovative and sustainable solutions across coastal protection, road stabilisation, erosion control, oil & gas infrastructure, and building construction. With over 28 years of proven expertise, our focus is on precision, partnership, and resilience — ensuring every project is completed on time, within budget, and to global standards. Together with our clients, we are shaping infrastructure that serves communities today and endures for generations.

B. Smith
Managing Director / Chief Executive Officer

A hybrid leader: combining boots-on-ground technical mastery with boardroom-level strategic acumen, he holds a first degree in Quantity Surveying and is an outstanding practitioner in the field. He brings a wealth of expertise in project design, estimation, production planning, logistics, and risk management. His leadership style blends **practical engineering insight** with **strategic vision**, enabling Baytree to execute projects that are both technically sound and commercially viable.
Bisola is an alumnus of the **Lagos Business School's Senior Management Program (SMP64)** and holds multiple technical certifications, including:

- Advanced Erosion Control Techniques (2012)
- Design Techniques for StreamBank Protection – Colorado State University/River Research & Design Inc., USA (2014)
- Diploma in Operations Management (2015)
- ISO 14001/9001 – 2015 Implementation Certificate – Root Cause Analysis (2018)

Career Highlights:

- Over 28 years of leadership in marine, civil, and structural engineering, delivering complex, multi-million-dollar projects. Proven expertise in driving projects from acquisition through execution, achieving international quality standards while maintaining community and stakeholder engagement.

Leadership Team:

Chairman

Our Chairman is a seasoned technical authority with over 30 years of hands-on and strategic experience in the engineering and construction industry. His career spans the successful delivery of high-value projects across diverse infrastructure sectors. His deep technical insight, governance acumen, and commitment to engineering excellence guide Baytree's long-term vision and ensure that every project is executed to the highest standards.

Director (Civil Works & Road Construction)

With over 18 years of experience, this COREN-registered civil engineer has built a reputation for excellence in road construction and large-scale civil works. A trusted consultant for multiple high-profile infrastructure projects, he brings expertise in pavement design, structural engineering, and construction management, ensuring every project meets both functional and durability requirements.

Director (Buildings & Structures)

A COREN-registered civil engineer with over 20 years of experience in building and structural engineering, this director specialises in delivering innovative, safe, and efficient building solutions. His core expertise spans structural design, high-rise construction, and complex building rehabilitation, making him an invaluable asset in Baytree's integrated construction projects.

Our Credibility and Industry Standing

- Extensive Track Record – Decades of experience through leadership and heritage companies in delivering infrastructure projects of national importance.
- Technical Excellence – A highly skilled workforce, supported by modern equipment and innovative methods.
- Certified Quality – Commitment to international best practices through ISO standards and specialised certifications.
- Financial Stability – Capability to manage and deliver projects of varying size and complexity without compromising quality or timelines.

Recognition and Awards

Baytree Engineering & Construction Limited's leadership has earned **national and international recognition** for technical expertise, industry contributions, and thought leadership. These milestones highlight our commitment to **continuous learning, innovation, and industry engagement**.

Awards and Industry Recognition

- **Service & Contribution Award – Nigerian Institution of Civil Engineers (NICE)**
- *Warri Chapter, 2014 Annual Seminar* – Presented to **our leadership** in recognition of exceptional service and contributions to the civil engineering profession in Nigeria.

Speaking Engagements & Thought Leadership

- **Speaker – West Africa Property Investment Summit (WAPI)**

Delivered industry insights on infrastructure development and sustainable engineering solutions to a regional audience of investors, developers, and policymakers.

- Speaker/Presenter at the Delta State Ministry of Works retreat in Abraka, 2018.

International Conferences and Training

- **World of Concrete Convention – USA (Annual Event):** Participated as part of Baytree's commitment to engaging with global leaders across geosynthetics, construction, soil stabilisation, and landscaping—ensuring we stay at the forefront of innovations and best practices shaping these industries.
- **Specialised Training – Geocells, Manufacturing and Their Applications:** Completed advanced technical training with GeoProducts in 2017 and 2023, enhancing our capacity to deploy cutting-edge ground stabilisation solutions
- **Training – Flint USA (flintusa.net) –** Training on geocoastal tubes and marine mattress production/installation on the international stage, sharing expertise on engineering best practices and innovations.

Our Credibility and Industry Standing

- **Extensive Track Record** – Decades of experience through leadership and heritage companies in delivering infrastructure projects of national importance.
- **Technical Excellence** – A highly skilled workforce, supported by modern equipment and innovative methods.
- **Certified Quality** – Commitment to international best practices through ISO standards and specialised certifications.
- **Financial Stability** – Capability to manage and deliver projects of varying size and complexity without compromising quality or timelines.

PREVIOUS PROJECTS AND
Case Studies

At Baytree Engineering & Construction Limited, our track record speaks for itself. We have consistently delivered complex infrastructure projects that combine **technical precision**, **innovative engineering**, and **environmentally responsible solutions**. Below are selected case studies showcasing our capacity to deliver on time, on budget, and to the highest quality standards.

Case Study 1: Bridge Pier Abutment Rehabilitation – Uzere, Isoko, Delta State

Client: Delta State Government / DESOPADEC
Sector: Public Works & Civil Engineering
Project Value: Confidential
Status: Completed

Project Overview

The bridge at Uzere, Isoko, is a vital transportation link connecting communities and facilitating the movement of goods and people. A critical section of the pier abutment had collapsed, rendering the bridge impassable and disrupting socio-economic activities. The cause was progressive erosion and structural degradation exacerbated by high seasonal water flow and inadequate initial protection measures.

BTE's Solution

Baytree Engineering & Construction Limited deployed an integrated structural and geotechnical solution, ensuring both immediate restoration and long-term resilience:

- **Gabion Systems** – Installed to stabilise and armour the affected section, offering flexibility and durability against hydraulic forces.
- **Terraforce Articulated Concrete Block Mattress (Terracrete)** – Used to protect against scour and erosion, providing a robust yet adaptable slope protection system.
- **Geotextile Underlay** – Laid beneath gabions and ACBM to prevent soil migration, enhance drainage, and extend service life.

Execution Highlights

- **Rapid Mobilisation** – Emergency response enabled swift site assessment and commencement of works.
- **Minimal Disruption** – Phased construction allowed partial access for essential movement during works.
- **Integrated Protection** – Combined systems provided multi-layered defence against future hydraulic impact.

Outcome & Impact

- Restored full access and functionality of the bridge within a short timeline.
- Provided a sustainable, low-maintenance structural solution designed to withstand seasonal flood impacts.
- Reinforced community confidence in infrastructure safety and resilience.

PREVIOUS PROJECTS AND
Case Studies

Case Study 2: Soil Stabilisation for NGIC Pipelines – Multiple Locations

Client: Nigerian Gas Infrastructure Company (NGIC)
Sector: Civil & Geotechnical Works / Oil & Gas Infrastructure
Project Value: Confidential
Status: Completed

Project Overview

The Nigerian Gas Infrastructure Company (NGIC) operates an extensive network of pipelines critical to the nation's energy supply. Several segments of these pipelines in Edo and Kogi States faced soil instability, exposing them to risks of differential settlement, erosion, and loss of cover. Left unaddressed, these conditions threatened operational safety, service continuity, and environmental integrity.

Locations:

- **Edo State:** Agbede (1), Fugar (1), Uromi (2)
- **Kogi State:** Okpangede (2), Power Station (1)

BTE's Solution

Baytree Engineering & Construction Limited implemented tailored soil stabilisation and erosion control measures at each site, ensuring long-term pipeline integrity:

- **Geotextile Layers** – To improve soil separation, filtration, and prevent fine soil migration.
- **Geocells & Geogrids** – To distribute loads, prevent subsidence, and reinforce slopes around exposed sections.
- **Gabion and Riprap Protection** – Where hydraulic action posed additional threats.
- **Re-profiling & Compaction** – To restore pipeline cover and ensure stable support.

Execution Highlights

- **Multi-Site Coordination** – Works executed simultaneously in different states, ensuring project efficiency.
- **Minimal Downtime** – All interventions were performed without disrupting gas transmission.
- **Rapid Deployment** – Emergency mobilisation to prevent further erosion during peak rainfall.

Outcome & Impact

- Restored structural stability of all affected pipeline segments.
- Reduced future maintenance and emergency intervention costs.
- Enhanced safety and operational reliability for NGIC's critical infrastructure.

PREVIOUS PROJECTS AND
Case Studies

Case Study 3: Coastal Protection Works – Okpoama, Brass, Bayelsa State (2024–2025)

Client: Regional Authorities
Sector: Marine & Coastal Infrastructure
Project Value: Confidential
Status: Completed (Phase 1) December 2024

Project Overview:

Between August and December 2024, Phase I of a major coastal protection initiative was executed in Okpoama, an area experiencing severe erosion. The solution combined dune core construction for immediate shoreline defence with offshore breakwaters designed to dissipate wave energy and encourage natural sedimentation.

Key Technical Highlights:

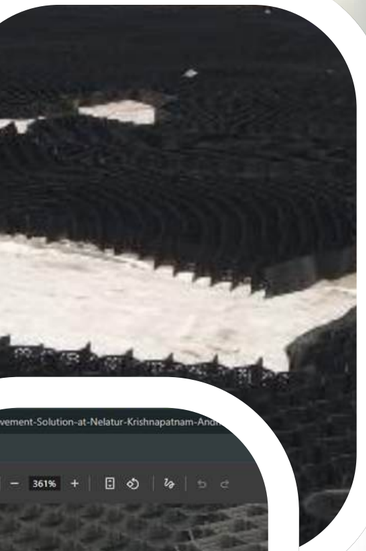
- **Geotextile Tubes:** Engineered to form the core of dune and breakwater structures, providing durable resistance to wave impact.
- **Concrete/Articulated Block Mattresses:** Protective armouring to extend structure lifespan and prevent damage..
- **Sediment Accretion:** Significant sand accumulation reshaped the beach profile within one month, with accretion averaging over 1 meter.
- **Community Access Maintained:** The design allowed fishing and local economic activities to continue uninterrupted.

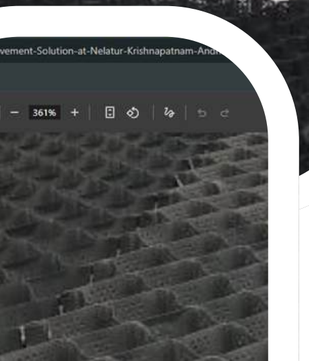
Outcome & Impact:

- Achieved **immediate erosion** control while promoting natural beach regeneration.
- Delivered a sustainable, low-maintenance solution with minimal environmental impact..
- Strengthened community resilience against climate-induced coastal challenges.

Project Leadership:

- Phase I was successfully executed under strong project direction, with oversight of execution, business development, and financial management, demonstrating effective leadership in complex, environmentally sensitive marine works.





PREVIOUS PROJECTS AND Case Studies

Case Study 4: Ground Improvement at Nelatur, Krishnapatnam, Andhra Pradesh (2020)

Sector: Integrated Construction Solutions / Ground Stabilisation
Client: Andhra Pradesh Power Development Company Limited (APPDCL)
Scope: Raise the ash pond bund of the Sri Damodaram Sanjeevaiah Thermal Power Plant by an additional 5 meters atop loose fly ash with very low bearing capacity.

Engineering Challenge:

The fly ash foundation posed serious risks of uneven settlement and structural instability due to its low dry density and non-plastic nature.

Baytree's solution (in collaboration with Techfab India):

- **Techcell Geocell:** A 3D honeycomb confinement system made of high-density polyethene, designed to expand and disperse load over a wider area—minimising pressure on the weak ash base.
- **TechGrid Biaxial Geogrid:** Positioned below the geocell, this added reinforcement ensured uniform load distribution and reduced differential settlements.
- **Nonwoven Geotextile (T200):** Served as a separator and filter, improving subgrade load transfer and soil stability.

Implementation Highlights:

- Installation was straightforward, with no need for highly skilled labour, facilitating efficiency in diverse field conditions.
- The geocell system improved shear strength, drainage, load capacity, and significantly reduced settlement risk under heavy loads.

Outcome & Impact:

- The upgraded bund achieved the **required structural integrity and bearing capacity**.
- Enhanced **operational safety and long-term reliability**, supporting thermal power infrastructure growth.
- Client approval was smooth; Techfab India's quality control protocols ensured that the factory and materials passed rigorous inspections before deployment.

BAYREE
ENGINEERING & CONSTRUCTION

PREVIOUS PROJECTS AND Case Studies

Case Study 4: River Bank Protection and Land Reclamation at Katol, Gujarat (2020)

Sector: Public Works & Civil Engineering / River Training Works
Client: INEOS Styrolution India Ltd.
Scope: Stabilisation and protection of a 184 m stretch of Orsang River bank with heights up to 9.6 m, plus land reclamation adjacent to a manufacturing plant.

Challenge: Severe erosion from monsoon flooding in a high-flow river zone, threatening industrial assets and causing annual land loss.

Solution (in partnership with Techfab India):

- **Reinforced Soil Wall with Gabion Fascia**, using **TechGrid Geogrid** for reinforcement and **Nonwoven Geotextile** for filtration.
- **Gabion Mattresses** as a launching apron to protect against scouring.
- Permeable gabion design to relieve hydrostatic pressure and allow vegetation growth for long-term stability.

Outcome:

- Provided a **flexible, eco-friendly alternative** to conventional RCC retaining walls.
- Stabilised the river bank while reclaiming land for industrial use.
- Reduced maintenance needs and integrated naturally into the environment.

Our Multi-Sector Capability in Action

These projects underscore Baytree's ability to:

- Deliver **turnkey marine, civil, and structural solutions** in challenging environments.
- Integrate advanced geosynthetic technologies through OEM partnerships with **Techfab India**, **Inland Tarp & Liner**, and **Terraforce**.
- Balance **engineering excellence** with **environmental responsibility** for sustainable, long-lasting results.

BAYTREE
ENGINEERING & CONSTRUCTION

Portfolio Strength Across Divisions

Baytree Engineering & Construction Limited operates across three core business divisions, enabling us to deliver specialised solutions for diverse infrastructure challenges. Our multi-sector portfolio demonstrates our ability to adapt, innovate, and execute at the highest level—regardless of project scale or complexity.

Division	Representative Project	Key Impact
Marine & Coastal Infrastructure	Coastal Protection Works – Okpoama, Brass, Bayelsa State	Restored over 1 meter of beach within one month of completion; safeguarded community assets from severe erosion while maintaining access for local fishing activities.
Public Works & Civil Engineering	River Bank Protection & Land Reclamation – Orsang River, Katol, Gujarat	Stabilized a 184 m riverbank, reclaimed industrial land, and implemented a flexible, eco-friendly alternative to RCC retaining walls.
Integrated Construction Solutions	Ground Improvement – Nelatur, Krishnapatnam, Andhra Pradesh	Enhanced load capacity and stability of a critical ash pond bund on weak soil, ensuring operational safety for a major thermal power facility.

Why This Matters for Our Clients:

- **Diverse Expertise:** From high-energy coastal zones to inland industrial facilities, our teams deliver tailored solutions.
- **Technology Integration:** Strategic partnerships with Techfab India, Inland Tarp & Liner, and Terraforce give us access to proven, world-class engineering products.
- **Sustainable Outcomes:** We prioritise eco-friendly, long-life solutions that minimise environmental impact while maximising project value.

Baytree's portfolio strength lies not just in the **breadth** of our capabilities, but in our ability to bring **specialised engineering knowledge** to each division—ensuring that every project benefits from the full depth of our technical resources and strategic partnerships.

Approach to the Project

At Baytree Engineering & Construction Limited, every project is approached with a structured, client-centric methodology that draws upon our multi-division expertise and proven track record. Our goal is not just to execute, but to exceed performance, quality, and sustainability benchmarks—delivering infrastructure that stands the test of time.

Precision planning

Innovation & Adaptability

Sustainable Delivery

Three core principles underpin our approach:

- **Precision Planning** – Harnessing technical expertise and strategic partnerships to develop solutions aligned with project specifications, site conditions, and client objectives.
- **Innovation & Adaptability** – Employing advanced products from our OEM partners and adapting global best practices to local realities.
- **Sustainable Delivery** – Embedding environmental responsibility in every stage to minimise impact and maximise long-term value.

BAYTREE
ENGINEERING & CONSTRUCTION

Approach to the Project

Step-by-Step Methodology

1. Comprehensive Needs Assessment

- Conduct detailed site investigations to understand environmental, technical, and logistical conditions.
- Engage with stakeholders to align project objectives with community and operational needs.

2. Customised Design & Engineering

- Select the most suitable solution pathway—drawing from our Marine & Coastal Infrastructure, Public Works, and Integrated Construction expertise.
- Develop engineering designs using a blend of structural performance modelling, environmental risk analysis, and value engineering.

3. Technology-Driven Implementation

- Deploy advanced geosynthetic solutions, structural systems, and construction technologies sourced from our global OEM partners (**Techfab India**, **Inland Tarp & Liner**, **Terraforce**).
- Utilise precision installation techniques to ensure longevity, safety, and low maintenance.

4. Quality & Safety Assurance

- Apply rigorous quality control protocols at every stage, from material sourcing to final inspection.
- Implement strict safety measures to protect workers, stakeholders, and surrounding environments.

5. Stakeholder Communication & Progress Reporting

- Maintain transparent, timely updates for the client and stakeholders to ensure alignment throughout the project lifecycle.

6. Handover & Long-Term Support

- Provide post-construction support, including maintenance guidance and performance monitoring where required.

Why Our Approach Works

Our project execution framework has been **proven across sectors**—from protecting high-energy shorelines in Bayelsa to reinforcing industrial embankments in Andhra Pradesh and stabilising riverbanks in Gujarat. By combining **site-specific innovation** with **decades of engineering experience**, we deliver results that are:

- **Technically robust** – built to withstand demanding operational and environmental conditions.
- **Cost-effective** – achieving optimal performance without unnecessary expenditure.
- **Sustainably engineered** – designed to last while preserving ecological balance.

At Baytree, our approach is not just about delivering a project—it's about creating **enduring value for our clients, communities, and the environment**.

Commitment to Quality, Safety, and Sustainability

At **Baytree Engineering & Construction Limited**, our reputation rests on **uncompromising excellence**. In our industry, success is measured not only by project completion, but by enduring performance, safety, and environmental responsibility.

Quality Assurance

Our principle is clear—do it right the first time, every time. We follow ISO-aligned processes and apply stringent material standards, with all products, including those from our OEM partners (**Techfab India, Inland Tarp** & **Liner**, **Terraforce**), tested to meet or exceed specifications. Multi-level inspections run from design to final handover, and value engineering ensures cost-effectiveness without compromising integrity.

Safety First

Safety is non-negotiable. We target zero lost-time incidents, with all staff receiving inductions, toolbox talks, and tailored training. Risks are identified, mitigated, and monitored under a proactive safety management system. Our protocols comply with Nigerian laws and global best practices.

Sustainability in Action

We embrace eco-friendly solutions, green technologies, and permeable materials. All major works undergo Environmental Impact Assessments to minimise disruption. Long-life, low-maintenance designs reduce future resource use, while projects are planned to benefit surrounding communities.

Through these commitments, Baytree ensures every project delivers lasting value—for clients, communities, and the environment.

Our Track Record Speaks

From the Okpoama Coastal Protection Project, restoring natural beaches, to the **Nelatur Bund Improvement**, stabilising power plant operations, and the Orsang Riverbank Reinforcement, integrating into the local ecosystem—our projects prove that quality, safety, and sustainability are the foundation of everything we deliver.

Financial Stability and Resources

Baytree Engineering & Construction Limited has the financial strength and resources to execute complex infrastructure projects efficiently and reliably. Our operational model maintains capital reserves, strong supplier relationships, and rapid access to specialised equipment—ensuring quick mobilisation and sustained project momentum.

Financial Strength

We are structured to fund both short- and long-term demands without disruption. With a history of delivering multi-million-dollar contracts through internal funding, strategic partnerships, and efficient cash flow management, we hold strong creditworthiness and adhere to recognised financial reporting standards for transparency.

Human Resources

Our leadership is headed by a CEO/MD with 28+ years' experience blending technical and executive expertise. A multidisciplinary workforce of engineers, project managers, and technicians undergoes continuous training, including Lagos Business School SMP64 alumni development.

Equipment and Material Resources

We have direct access to cutting-edge equipment for marine works, geosynthetics, earthworks, and structural construction. OEM partnerships with Techfab India, Inland Tarp & Liner, and Terraforce ensure consistent product quality and availability. Our fleet and specialised tools enable efficient, on-site execution.

Supply Chain Resilience

A vetted global and domestic vendor network supports material quality and on-time delivery. With proven logistical expertise, we coordinate complex procurement and deliveries even in challenging terrains or high-risk environments.

Proven Capacity to Deliver

From coastal defence in Bayelsa to ground improvements in Andhra Pradesh and riverbank reclamation in Gujarat, Baytree proves its resources, financing, and readiness to deliver globally, on time and within budget.

Risk Management and Contingency Plans

Baytree Engineering & Construction Limited applies a proactive, systematic risk management approach, ensuring potential challenges are identified, assessed, and mitigated before they affect timelines, budgets, or quality. Our philosophy is simple: anticipate, plan, and act—delivering seamless project outcomes even in complex conditions.

Our Risk Management Framework

Risk management is integrated throughout the project lifecycle:

- **Early Identification:** Comprehensive technical, environmental, financial, and operational assessments at inception.
- Quantitative & Qualitative Analysis: Risks prioritised by likelihood, severity, and potential impact, enabling efficient resource allocation.
- **Mitigation Planning:** Targeted prevention and mitigation strategies to minimise probability and consequences.
- **Continuous Monitoring:** Real-time tracking of progress, site conditions, and market factors for swift corrective action.

Key Risk Categories and Mitigation

- **Technical & Engineering:** Rigorous design reviews, peer checks, OEM-certified installations.
- **Environmental:** Seasonal scheduling, EIAs, and resilient engineering designs.
- **Supply Chain:** Multi-supplier sourcing, OEM partnerships (Techfab India, Inland Tarp & Liner, Terraforce), advanced procurement.
- **Financial:** Detailed cost planning, contingency reserves, milestone-based disbursements.
- **Safety:** HSE compliance, safety training, regular audits.
- **Stakeholder:** Transparent communication, early engagement, continuous reporting.

Through structured risk control and robust contingency planning, Baytree consistently safeguards project delivery—on time, within budget, and to global quality standards.

Risk Management and Contingency Plans

Contingency Planning

We recognise that even the most well-managed projects can face unexpected challenges. Our contingency plans ensure we remain agile and responsive:

- **Emergency Response Protocols:** Rapid mobilisation teams for structural, environmental, or operational incidents.
- **Redundancy in Resources:** Backup equipment, alternate suppliers, and additional workforce capacity ready to deploy if needed.
- **Flexible Scheduling:** Ability to re-sequence activities to maintain progress during unforeseen delays.
- **Financial Buffering:** Contingency funds built into project budgets to address cost variations without affecting overall delivery.

Proven Resilience in Action

Our past projects—such as the Okpoama Coastal Protection, which faced unpredictable wave energy conditions, and the Orsang Riverbank Reclamation, delivered in a flood-prone zone—demonstrate our ability to adapt quickly while maintaining quality and safety. This adaptability is a core reason clients trust Baytree for projects where risk is high, but failure is not an option.

Competitive Advantage

Baytree Engineering & Construction Limited distinguishes itself in a competitive engineering and construction market by combining specialist technical expertise, world-class product integration, and a client-first execution philosophy. Our edge is built on five key pillars:

1. Multi-Division Expertise with Proven Results

Operating across **Marine & Coastal Infrastructure**, **Public Works & Civil Engineering**, and **Integrated Construction Solutions**, we deliver tailored solutions to diverse and complex challenges. Our experience spans restoring eroded shorelines in high-energy marine zones, stabilising industrial embankments on weak soils, and reinforcing riverbanks in flood-prone areas while reclaiming land.

2. **Strategic OEM Partnerships for World-Class Solutions**

Exclusive and preferred partnerships with **Techfab India**, **Inland Tarp & Liner**, and **Terraforce** give us access to globally recognised engineering products. These include Techfab India's TechMatt System—advanced pre-fabricated concrete mattress solutions for slope protection, scour prevention, and hydraulic erosion control—and proprietary geosynthetics and gabion technologies engineered for flexibility, drainage, and long-term durability. These alliances ensure technical superiority, supply chain resilience, and consistent material quality.

3. Innovation in Engineering and Delivery

We employ **advanced geotechnical** and **hydraulic solutions** that outperform traditional methods in speed, cost efficiency, and sustainability, such as geotextile tube and mattress systems, drainage-enabled linings, and flexible modular construction that adapts to dynamic forces and settlements.

4. Sustainability as a Core Value

Our solutions reduce rock quarrying by up to 40%, integrate vegetation for slope stabilisation, and deliver low-maintenance, long-life infrastructure that minimises environmental impact and resource use.

5. Leadership That Delivers

Led by **Smith Bisola A.**, a seasoned leader with over 28 years of engineering and executive experience, including Lagos Business School training, Baytree combines on-site technical mastery with strategic boardroom insight for flawless execution.

The Result: Clients gain a strategic partner with global-standard solutions, proven delivery in challenging environments, innovation-driven designs, and the capacity to execute high-value projects without compromise.

Financial Proposal and Pricing

At Baytree Engineering & Construction Limited, our pricing framework is built on **transparency, value delivery, and cost-effectiveness**. We understand that in high-stakes infrastructure projects, clients need more than a competitive price—they need assurance that every investment translates into **measurable quality and performance**.

Our Pricing Philosophy

1. **Value Over Lowest Price** – We focus on providing solutions that deliver the best *total lifecycle value*, not just the lowest initial cost. Our designs and materials are selected to minimise future maintenance, extend service life, and reduce operational disruptions.
2. **Tailored to Project Needs** – Every project is unique. We prepare detailed cost breakdowns based on specific scope, site conditions, and performance requirements to ensure clients pay for **precisely what they need—nothing more, nothing less**.
3. **Transparency in Costing** – All pricing elements, from materials and labour to equipment and logistics, are clearly itemised, giving clients full visibility. There are no hidden charges or unplanned cost inflations.
4. **Efficiency Through Innovation** – By integrating advanced engineering solutions, such as **geosynthetic technologies from our OEM partners**, we achieve efficiencies that allow us to offer high-quality solutions at competitive rates.
5.

What Clients Gain from Our Financial Model

- **Long-Term Savings** – Reduced maintenance costs through durable, low-maintenance systems.
- **Risk Mitigation** – Contingency planning and resilient supply chains ensure timely delivery without costly delays.
- **Performance Assurance** – High-grade materials and strict quality controls safeguard the investment for decades.
- **Optimised Resource Allocation** – Our engineering-led planning ensures every resource—time, manpower, and capital—is maximised for client benefit.

Our financial proposals are designed not just to win tenders, but to **build lasting trust**—ensuring that every dollar invested in Baytree delivers maximum return in safety, quality, and operational performance.

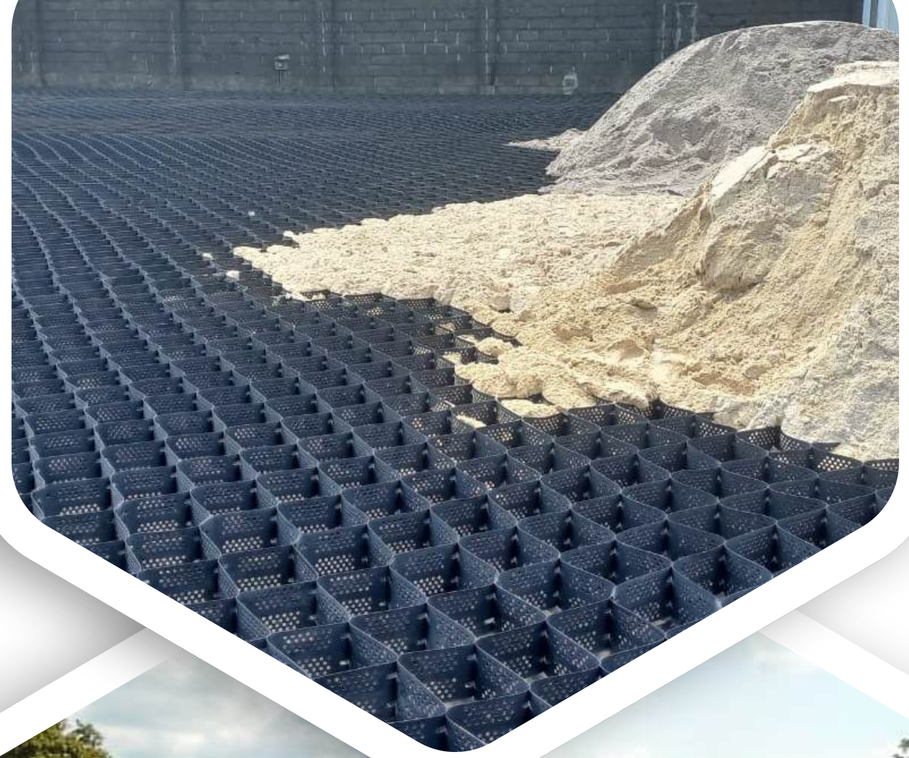
BAYTREE
ENGINEERING & CONSTRUCTION

August 2025

Part Two

Product Catalogue/Brochure

Engineered Products for Durable, Sustainable Infrastructure



Presented by
Baytree Engineering & Construction Limited

BAYTREE
ENGINEERING & CONSTRUCTION

Website:
baytreeng.com

Our Products

Geosynthetic Solutions

- **Geotextiles** – Engineered permeable fabrics used to separate, filter, reinforce, protect, or drain soil. They are essential in road construction, retaining structures, embankments, and erosion control, allowing water to pass while preventing soil migration.

- **Geocells** – Three-dimensional honeycomb-like cellular confinement systems made from high-density polyethene. When filled with soil, aggregate, or concrete, they create a stable, load-distributing platform. Geocells are ideal for slope protection, load support in weak soils, and erosion prevention in channels and embankments.

.

- **Geogrids** – High-strength polymer grids designed to reinforce and stabilise soil by interlocking with the aggregate or fill material. They significantly improve load-bearing capacity and reduce deformation, making them suitable for base reinforcement in roads, railways, retaining walls, and embankments.

Together, these geosynthetics provide durable, cost-effective solutions for infrastructure stability, extending service life and reducing maintenance costs.

Our Products

Geosynthetic Solutions

- **Geotubes & Geobags** – Large and small geotextile containers engineered for a variety of hydraulic and geotechnical applications. They are widely used in dewatering dredged materials, creating shoreline barriers, and providing erosion control. When filled with sand, soil, or dredged material, they form robust, stable structures capable of withstanding wave action and currents.

- **Geomembranes** – Impermeable synthetic liners designed for containment and environmental protection. Commonly used in water reservoirs, canals, landfill caps, tailing ponds, and industrial containment systems, geomembranes prevent fluid migration and protect surrounding soil and groundwater from contamination.

- **Sack Bags** – Compact, durable geotextile bags used for sediment control, shoreline defence, and oil and gas operations. Easily filled and deployed, they provide temporary or permanent protection against erosion, stabilise embankments, and contain hazardous materials in spill response scenarios.

These solutions offer versatility, durability, and cost-effectiveness for both temporary works and long-term infrastructure protection.

Our Products

BAYREE
ENGINEERING & CONSTRUCTION

Hydraulic and Erosion Control Systems

TechMatt Fabric Form Concrete Mattress Systems: Pre-fabricated, high-strength geotextile formwork for slope protection, scour prevention, and hydraulic erosion control. Available in:

- **Uniform Section Concrete Mattress** – Continuous surface protection for slopes and channels.
- **Filter Point Concrete Mattress** – Drainage-enabled to relieve subsoil pressure and prevent soil migration.
- **Articulated Block Mattress** – Flexible concrete block system for coastal revetments, tidal zones, bridge piers, and river training works.

Our Products

BAYREE
ENGINEERING & CONSTRUCTION

Masonry and Hardscaping Units

- **Pavers:** High-quality paving stones for roads, walkways, and public spaces.
- **Hardscaping Stones:** Natural stone products for landscaping and architectural finishes.
- **Retaining Wall Blocks:** Modular systems for earth retention, slope stabilisation, and decorative walls.
- **Decorative Blocks:** Architectural masonry units for visual enhancement in landscaping projects.

Our Products

Infrastructure Solutions

- **Erosion Control Systems:** Engineered approaches to protect slopes, shorelines, and embankments.
- **Stormwater Management Systems:** Channels, linings, and barriers to control runoff and reduce flood risk.
- **Canal and Channel Linings:** Durable, low-maintenance protection for irrigation and drainage networks.

Advanced Fabric Form Grout Mattress System

BAYREE
ENGINEERING & CONSTRUCTION

TECHMATT

- Long-Term Hydraulic applications
- Durable
- Versatile
- Engineered for Performance

BAYREE
ENGINEERING & CONSTRUCTION

Tech Matt

TechMatt US

TechMatt is our partner's (TechFab India) advanced pre-fabricated fabric form concrete mattress system, developed to offer a robust, adaptable, and sustainable solution for slope protection, scour prevention, and hydraulic erosion control. Manufactured using high-strength geotextile formwork and in-situ pumped concrete or grout, TechMatt provides a seamless lining system suitable for a wide range of hydraulic and geotechnical challenges.

Engineered for Suitability and performance:

- Forms a monolithic, continuous surface for protecting slopes and channels.
- Ideal for lining canals, embankments, and water
- conveyance structures.
- Offers excellent resistance to surface erosion and
- ensures slope integrity.

BAYREE
ENGINEERING & CONSTRUCTION

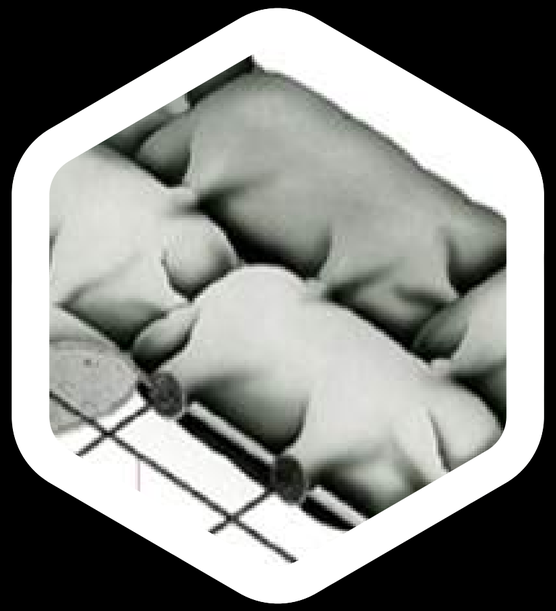
Tech Matt

★★★★★

TechMatt FP

(Filter Point Concrete Mattress)

- Engineered with integrated drainage ports to relieve subsoil water pressure.
- Minimises risks of hydrostatic uplift, soil migration, and piping.
- Ideal for high water table areas, riverbanks, and seepage-prone slopes.



★★★★★

TechMatt AB

(Articulated Block Mattress)

- Consists of concrete blocks linked by geotextile joints forming an articulated matrix.
- Offers high flexibility to accommodate settlements and hydraulic impacts.
- Best suited for coastal revetments, tidal zones, bridge piers, and river training works.
- Optional inclusion of tensile cables for enhanced anchorage in dynamic zones.

Our Partners

1 TECHFAB India

TECHFAB India: A leading manufacturer and supplier of world-class geosynthetic and geotechnical products, **TECHFAB India** is renowned for its innovation, quality, and engineering excellence.
Their solutions, ranging from geotextiles, geogrids, geocells, and geotubes to advanced systems like the TechMatt concrete mattress, are trusted globally for infrastructure, environmental, and hydraulic applications.
With in-house design, manufacturing, and technical support capabilities, TECHFAB India ensures products meet the highest international standards.

Our partnership enables **Baytree Engineering & Construction Limited** to integrate cutting-edge, proven technologies into our projects, delivering durable, sustainable, and cost-effective solutions tailored to the unique challenges of marine, civil, and geotechnical engineering.

Website: baytreeng.com

Our Partners

2 Inland Tarp and Liner

Inland Tarp & Liner: A respected U.S.-based manufacturer, **Inland Tarp & Liner** specialises in high-performance industrial fabrics and custom-fabricated containment solutions. Their products include geomembranes, liners, tarpaulins, and protective covers designed for demanding applications in agriculture, construction, mining, oil and gas, and environmental protection.
Known for durability, precision fabrication, and rapid delivery, **Inland Tarp & Liner** provides solutions that perform in the most challenging conditions.

Through our partnership, **Baytree Engineering & Construction Limited** gains access to premium-quality containment and lining systems, enabling us to deliver reliable, long-life infrastructure solutions that meet strict environmental and operational standards in marine, civil, and industrial projects.

Website: baytreeng.com

BAYTREE
ENGINEERING & CONSTRUCTION

Our Partners

3 Terraforce

Terraforce: A globally recognised leader in modular retaining wall systems, **Terraforce** designs and supplies eco-friendly, durable, and versatile concrete blocks for earth retention, slope stabilisation, and landscaping.
With over 45 years of proven performance, **Terraforce** blocks feature a hollow-core, interlocking design that allows for curved, angled, or vertical walls without complex fittings.
They promote soil stabilisation while enabling vegetation growth, blending functionality with aesthetic appeal. Widely used in commercial, residential, and infrastructure projects, Terraforce solutions are engineered for longevity and adaptability.

Our partnership allows **Baytree Engineering & Construction Limited** to deliver retaining wall systems that are cost-effective, sustainable, and architecturally pleasing.

Website: baytreeng.com

BAYTREE
ENGINEERING & CONSTRUCTION

BAYTREE
ENGINEERING & CONSTRUCTION

Partner With Us

Baytree Engineering & Construction Limited is ready to **partner with you** to deliver infrastructure solutions that combine **technical excellence, sustainability, and cost efficiency**.
Whether the challenge lies in **protecting coastlines from relentless wave energy**, **stabilising critical industrial structures**, or **delivering complex public works**, our team has the **expertise, resources, and track record** to execute with confidence.

We invite you to:

- **Engage with our technical team** for a detailed discussion on how we can tailor our solutions to your project needs.
- **Review our past performance** to see first-hand the scale, quality, and innovation we bring to every contract.
- **Collaborate with a partner who delivers**—on time, within budget, and beyond expectation.

Let's build not just structures, but lasting value: together.

Contact us at:

Address
Head Office: Plot 6 Idowu Olaitan Street, Gbagada Phase 2, Lagos

Phone
+234-805-777-7724 | +234 803 945 8755 | +234 906 000 1606

Email
info@baytreeng.com

Website
www.baytreeng.com

IG
baytreeengineering